DATA ITEM DESCRIPTION

Title: DYNAMIC RESPONSE ANALYSIS AND TEST REPORT

Number: DI-SESS-82489
Approval Date: 20250709
AMSC Number: F10575
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Preparing Activity: 11 (AFLCMC/EZFS)
Project Number: SESS-2025-014
Applicable Forms: N/A

Use/Relationship: The Dynamic Response Analysis and Test Report describes the ground and flight analyses and tests used to characterize the system elastic response to dynamic load inputs. It provides a means for verifying or improving the dynamic loads analysis and fatigue loads spectrum for the air vehicle, missile, or helicopter weapon system. Results are incorporated into the fatigue test program to determine service life for actual operational usage.

a. This Data Item Description (DID) is applicable to any contract for which a dynamic response test is required.

b. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in section 5 of MIL-STD-1530, *Aircraft Structural Integrity Program (ASIP)* (copies of this document are available online at https://quicksearch.dla.mil).

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The Dynamic Response Analysis and Test Report shall be in the contractor's format.

3. Content. The Dynamic Response Analysis and Test Report shall describe the contractor's analyses and testing activities associated with the design for dynamic load inputs. The report shall contain the results and findings used to validate or correct the initial dynamic response analyses.

a. The report shall be based on the interpretation and evaluation of the dynamic response analyses and tests and shall contain the following, as applicable:

(1) Descriptions and photographs of the system, fixturing, rigging, ground equipment, instrumentation, component(s) under test, and the configurations including fuel states, store loadings, weights, centers-of-gravity, internal payloads, and simulated sling loads.

(2) Descriptions of the flight or ground handling conditions that result in significant dynamic responses of the structure, and the effects of these conditions on service life.

(3) Descriptions of the test procedures, test conditions, test methods, and data acquisition systems, including a copy of the final version of the test plan.

(4) Descriptions of the methodologies used for selecting test instrumentation types, test instrumentation measurement capabilities, and test measurement locations.

(5) Specifications for all test instrumentation.

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(6) Description of the methods used for mounting instrumentation.

(7) Descriptions of the methods used for the reduction and analysis of the test data.

(8) Copy of the test matrix as planned to be executed.

(9) Defined measurement units including axis labels of plots and figures.

(10) Plots of the transfer functions versus frequency for vibration sensors.

(11) Descriptions and time history plots of critical test load cases.

(12) Copy of the test log as executed.

(13) Results from both discrete and power spectral density methods of analysis and a comparison of the load inputs, structural responses, and transfer functions with the dynamic loads analysis.

(14) The transfer functions between the internal structural loads and accelerations at various locations throughout the airframe and the gust, landing, taxi, takeoff, towing, buffet, aerial refueling, catapult, arrestment, gun fire, and store ejection load inputs.

(15) The transfer functions between vertical center of gravity accelerations and the load frequency distribution for the range of gross weights, system configurations, speeds, angles of attack, angles of sideslip, and altitudes being flown in service.

End of DI-SESS-82489

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